

# Isomorph-Eval: Structurally Equivalent Benchmarks Reveal That Annotation-Chain Replay Produces False Contamination Signals

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May 2026

## Abstract

We present **Isomorph-Eval**, a framework for generating structurally equivalent benchmark variants via  $\tau$ -*isomorphism*, a graph-theoretic definition that preserves operation types, dependency structure, and complexity weights while ensuring complete surface novelty. The framework’s four-stage pipeline (Parser  $\rightarrow$  Mutator  $\rightarrow$  Generator  $\rightarrow$  Verifier) produces unlimited verified variants of existing benchmark items with automatically computed answers.

Applying Isomorph-Eval to GSM8K, we discover that *annotation-chain replay*—the standard approach to computing variant answers by substituting values into the original solution chain—produces systematically wrong answers in 41% of items. We identify eight bug classes (value conflation, phantom leaf values, positional replacement collisions, magnitude inflation, answer invariance, partial name replacement, node reuse, and inconsistent entity replacement) that create false contamination signals: all five evaluated models appeared uniformly contaminated ( $\Delta \approx +0.42$ ) on the unverified data, including a comparison model whose training data composition has not been fully disclosed.

After independent verification and pipeline repair, the signal collapses. On 63 entity-clean verified items, four of five models show equivalent performance on originals and variants ( $\Delta_{\text{iso}} \in [+0.036, +0.047]$ , all  $p > 0.05$ ), while Llama 3.1 8B shows a mild gap ( $\Delta_{\text{iso}} = +0.114$ ) consistent with known numeric perturbation sensitivity. We argue that independent answer and entity verification are necessary for any benchmark variant generation framework, and release the complete pipeline as open-source software.

## 1 Introduction

Benchmark contamination—the presence of test items in training data—is widely suspected of inflating LLM performance on standard evaluations Zhang et al. [2024], Dekoninck et al. [2024], Zhao et al. [2025]. The standard remedy is to generate novel benchmark variants that test the same reasoning while eliminating memorization advantages. But a critical question has received insufficient attention: *how do we know the variant answers are correct?*

We present **Isomorph-Eval**, a framework for generating structurally equivalent benchmark variants via  $\tau$ -isomorphism, a graph-theoretic definition of structural equivalence. In applying the framework to GSM8K Cobbe et al. [2021], we discovered that the standard approach to computing variant answers—annotation-chain replay, in which new values are substituted into the original solution chain—produces systematically wrong answers in 41% of items. These errors created an apparent contamination signal of  $\Delta \approx +0.42$  that was *uniform across all models*, including a comparison model whose training data composition has not been fully

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disclosed. After independent verification and pipeline repair, the signal collapsed to  $\Delta \approx +0.04-0.11$ .

This finding motivates a reframing: the primary contribution of this work is not contamination detection, but rather the identification of eight bug classes in variant generation pipelines that produce false contamination signals, and the development of verification methods that prevent them.

**Background.** Prior work on benchmark contamination detection falls into two categories. *Parallel benchmark* approaches construct entirely new test sets: GSM1K Zhang et al. [2024] reported 8–13% performance drops for open-source models on a manually curated GSM8K parallel; MMLU-CF Zhao et al. [2025] found a 14.6-point drop for GPT-4o on a contamination-free MMLU variant. *Statistical detection* approaches such as ConStat Dekoninck et al. [2024] and PaCoST use performance gaps on rephrased items as contamination evidence. A complementary line of work has established that benchmark scores are also distorted by aggregation methodology: our prior Evaluation Failure Scaling Law (EFSL) Kang [2026] showed that simple averaging degrades to  $\rho = 0.24$  under the joint effect of data sparsity and item difficulty heterogeneity, while PSN-IRT Zhou et al. [2026] and MedIRT Luo et al. [2025] demonstrated that IRT-based rankings are more robust.

All of these approaches assume that the alternative evaluation—whether a parallel benchmark or a rephrased variant—has correct answers. None systematically verify this assumption. GSM-Symbolic Mirzadeh et al. [2025] comes closest, substituting symbolic values into GSM8K solution templates. VarBench reports that Llama 3 Instruct 8B drops from 75.8% on GSM8K to 39.4% on GSM+ (48.3% relative drop) under dynamic variable perturbation Qian et al. [2024], underscoring how difficult it is to separate perturbation sensitivity from contamination.

**$\tau$ -Isomorphism.** We formalize the concept of structural equivalence as  $\tau$ -*isomorphism* (Section 2.1): two benchmark items are  $\tau$ -isomorphic if there exists a graph isomorphism between their reasoning graphs that preserves operation types, dependency structure, and complexity weights. Unlike semantic similarity heuristics used by prior rephrasing approaches,  $\tau$ -isomorphism is a deterministic, verifiable structural condition. The *Isomorphic Engine* (Section 2.3) implements a four-stage pipeline (Parser  $\rightarrow$  Mutator  $\rightarrow$  Generator  $\rightarrow$  Verifier) that automatically produces verified  $\tau$ -isomorphic variants with answers computed by forward graph execution.

**The cautionary finding.** Applying the Isomorphic Engine to a stratified sample of 100 GSM8K items, we found that annotation-chain replay—the mechanism that computes variant answers by substituting new values into the original solution annotations—fails in three systematic ways:

1. **Value conflation:** the same numeric value serves multiple semantic roles in the problem (e.g., “5 cars” and “\$5 each”), but the solution chain merges them into a single graph node; mutating the node changes both uses in the graph but only one in the text.
2. **Phantom leaf values:** the solution chain introduces intermediate results as leaf nodes (e.g.,  $\langle 40=40 \rangle$ ) that do not appear in the question text; mutating these phantom values produces wrong answers because the graph is structurally incomplete.
3. **Positional replacement collisions:** sequential text replacement causes new values to shadow old values of other nodes, producing text that does not match the graph computation.

These three failure modes, along with five additional bug classes (magnitude inflation, answer invariance, partial name replacement, node reuse, and inconsistent entity replacement), produced wrong or corrupted variants in 41 of 100 items. All five evaluated models—including a comparison model—appeared uniformly “contaminated” ( $\Delta \approx +0.42$ ) because they were penalized for

*correctly solving* variants whose expected answers were wrong.

**Corrected results.** After fixing the generation pipeline and independently verifying all variant answers via forward graph execution and filtering variants with inconsistent entity replacement, we evaluate five LLMs on 63 entity-clean verified items. Four of five models show no significant performance gap between originals and variants ( $\Delta_{\text{iso}} \in [+0.036, +0.047]$ , all  $p > 0.05$ ). The remaining model, Llama 3.1 8B, shows a mild gap ( $\Delta_{\text{iso}} = +0.114$ ,  $p = 0.003$ ), far smaller than the Llama 3 Instruct 8B drop reported by VarBench (75.8% on GSM8K to 39.4% on GSM+, 48.3% relative) Qian et al. [2024]. GPT-OSS 120B serves as a comparison model: while its training data is not fully disclosed, its open-weight status and STEM-focused training make GSM8K inclusion likely, so its small  $\Delta_{\text{iso}}$  ( $+0.036$ ,  $p = 0.090$ ) suggests the  $\tau$ -isomorphic variants are not systematically harder rather than confirming absence of contamination.

### Contributions.

1. We formalize  $\tau$ -*isomorphism* as a verifiable graph-theoretic condition for benchmark item equivalence (Section 2.1).
2. We design and release the *Isomorphic Engine*, a four-stage pipeline that produces unlimited verified variants with automatically computed answers and deterministic structural verification (Section 2.3).
3. We identify **eight bug classes** in annotation-chain replay that produce false contamination signals, and demonstrate that *independent answer verification via forward graph execution* is necessary for any benchmark variant generation framework (Section 4.2).
4. We provide empirical evidence that on verified data, current LLMs show robust reasoning transfer to novel numeric contexts, with remaining gaps consistent with perturbation sensitivity rather than contamination (Section 4).
5. We define  $\Delta_{\text{iso}}^{\text{IRT}}$ , a DIF-based metric for measuring perturbation sensitivity with IRT discrimination weighting (Section 2.4).

We release the complete pipeline as open-source software.<sup>1</sup>

## 2 The Isomorph-Eval Framework

We now present the three components of Isomorph-Eval: the formal definition of task isomorphism (§2.1), the generative engine (§2.3), and the perturbation sensitivity diagnostic (§2.4).

### 2.1 Item Decomposition and $\tau$ -Isomorphism

**Definition 1** (Item Decomposition). *An evaluation item  $i$  is a tuple  $(S_i, G_i, V_i)$  where:*

- $S_i$  is the surface representation: *the natural-language string presented to the model;*
- $G_i = (N_i, E_i, \tau_i, \phi_i)$  is the reasoning graph, *a directed acyclic graph where  $N_i$  is a finite set of computation nodes,  $E_i \subseteq N_i \times N_i$  is a set of directed dependency edges,  $\tau_i : N_i \rightarrow \mathcal{T}$  maps each node to an operation type from a finite taxonomy  $\mathcal{T}$ , and  $\phi_i : N_i \rightarrow \Phi$  assigns a complexity weight to each node;*
- $V_i : \mathcal{A} \rightarrow \{0, 1\}$  is the verification function *that determines the correctness of a candidate answer.*

The operation taxonomy  $\mathcal{T}$  is domain-specific. For mathematical reasoning (our primary domain), we define:

$$\mathcal{T}_{\text{math}} = \{\text{ASSIGN, ADD, SUBTRACT, MULTIPLY, DIVIDE, MODULO}\}. \quad (1)$$

<sup>1</sup>Code and data: <https://github.com/testofschool/isomorph-eval>

The complexity weight  $\phi(n) = (k, d, f, s)$  encodes the number of operands  $k$ , the digit class  $d$  (the maximum number of digits of any operand), a Boolean  $f$  indicating the presence of fractions, and a Boolean  $s$  indicating negative values. The digit class is the primary driver of arithmetic difficulty: multiplying two 3-digit numbers is substantially harder than multiplying two 1-digit numbers, even though both are instances of MULTIPLY.

Leaf nodes are ASSIGN operations with no incoming edges, representing the given quantities in the problem. Internal nodes represent computation steps. The answer is the value at a designated output node  $n^* \in N_i$ .

**Definition 2** ( $\tau$ -Isomorphism). *Two items  $i$  and  $i'$  are  $\tau$ -isomorphic, written  $i \cong_\tau i'$ , if and only if there exists a bijection  $\psi : N_i \rightarrow N_{i'}$  satisfying all of the following:*

- (i) **Structural preservation:**  $(n_a, n_b) \in E_i \iff (\psi(n_a), \psi(n_b)) \in E_{i'}$ .
- (ii) **Operation-type preservation:**  $\forall n \in N_i : \tau_i(n) = \tau_{i'}(\psi(n))$ .
- (iii) **Complexity preservation:**  $\forall n \in N_i : \phi_i(n) = \phi_{i'}(\psi(n))$ .
- (iv) **Semantic novelty:**  $\text{sim}(S_i, S_{i'}) < \delta$  for a threshold  $\delta$ , and  $S_{i'} \notin \mathcal{C}_{\text{web}}$ , where  $\mathcal{C}_{\text{web}}$  denotes the set of strings present in publicly crawled web corpora.

Conditions (i)–(iii) preserve the *cognitive structure* of the item: the same chain of reasoning steps, in the same dependency order, with the same quantitative complexity at each step. Condition (iv) reduces the risk of direct surface memorization. Note that web-novelty is necessary but not sufficient for guaranteeing absence from all training corpora, as training sets may include non-public or synthetic data. Unlike prior rephrasing approaches Dekoninck et al. [2024], Zhang et al. [2024] that assess “similar difficulty” via heuristic comparison,  $\tau$ -isomorphism is a *deterministic, verifiable* structural condition: given two reasoning graphs, one can check in polynomial time (via graph isomorphism algorithms such as VF2 Cordella et al. [2004]) whether  $\psi$  exists.

**Structural fingerprint.** For efficient pre-screening, we define a hash

$$h(G_i) = \text{SHA256}(|N_i|, |E_i|, \text{depth}(G_i), [\tau(n)]_{n \in \text{topo}(G_i)}, [\phi(n)]_{n \in \text{topo}(G_i)}) \quad (2)$$

computed over the topological ordering of nodes. Two items can be  $\tau$ -isomorphic only if  $h(G_i) = h(G_{i'})$ ; this provides  $O(1)$  rejection of non-isomorphic pairs before invoking the full VF2 check.

## 2.2 The Reasoning Structure Hypothesis

The utility of  $\tau$ -isomorphism for contamination detection rests on the following conjecture linking graph structure to IRT parameters.

**Conjecture 1** (Reasoning Structure Hypothesis). *Let item  $i$  have IRT parameters  $(a_i, b_i)$  estimated from a sufficiently large calibration panel of  $J$  systems. If  $i \cong_\tau i'$ , then:*

$$|b_i - b_{i'}| \leq \epsilon_b \quad \text{and} \quad |a_i - a_{i'}| \leq \epsilon_a \quad (3)$$

for tolerance bounds  $\epsilon_b, \epsilon_a$  that depend on the granularity of the operation taxonomy  $\mathcal{T}$  and the complexity weight space  $\Phi$ .

The intuition is that difficulty  $b$  in the 2PL model reflects the ability level at which 50% of systems succeed. If two items require the same chain of reasoning operations with the same structural dependencies and the same quantitative load at each node, then a system that cannot perform the required computation chain will fail both, and a system that can will succeed at both—regardless of whether the entities are named “Janet” or “Tomás” and whether the units are “eggs” or “honey jars.”

This is a conjecture, not an axiom: surface features *can* affect difficulty through secondary channels (unfamiliar vocabulary, numeric magnitude within digit class, culturally specific framing). The complexity weight  $\phi$  mitigates the dominant source of such leakage (digit class), and the empirical calibration protocol (Section 2.4) provides a falsifiability mechanism.

### 2.3 The Isomorphic Engine

The Isomorphic Engine is a four-stage pipeline that automatically generates verified  $\tau$ -isomorphic variants. Given an original item  $(S, G, V)$ , it produces a variant  $(S', G', V')$  such that  $G \cong_\tau G'$  and  $V'$  is automatically determined by forward execution of  $G'$ .

**Stage 1: Parser** ( $S \rightarrow G, V$ ). We employ a hybrid strategy. A rule-based annotation parser extracts the computation DAG from GSM8K’s «**expr=result**» annotations, achieving 57.2% coverage on the test set (755/1,319 problems) with zero LLM involvement. For the remaining items, we use LLM structured extraction via Pydantic-enforced schemas and the Anthropic tool-use API, with the LLM’s output validated by forward graph execution.<sup>2</sup>

**Stage 2: Mutator** ( $G \rightarrow G'$ ). The mutator operates exclusively on the *free dimensions* of the item: entity bindings (e.g., “Janet”  $\rightarrow$  “Tomás”), numeric leaf values (preserving digit class), units (“dollars”  $\rightarrow$  “euros”), and context theme. The *locked invariants*—node set, edge set, operation types  $\tau$ , and complexity weights  $\phi$ —are never modified. After mutation, the answer  $V'$  is recomputed by forward execution of  $G'$ , requiring no human annotation.

**Stage 3: Surface Generator** ( $G' \rightarrow S'$ ). We use a *template-first* architecture. A deterministic template containing all mathematical content is generated from  $G'$ , then optionally polished by an LLM. The LLM’s system prompt strictly forbids changing any number or logical relationship; its role is exclusively that of a surface realizer, not a reasoner. The prompt enforces six explicit constraints, and the output is post-checked for number preservation.

**Stage 4: Verifier** ( $S' \rightarrow G'' \cong_\tau G'$ ). The critical safety mechanism is *round-trip verification*: the generated surface text  $S'$  is re-parsed into a reasoning graph  $G''$  using the same parser as Stage 1, and  $G''$  is checked against  $G'$  for  $\tau$ -isomorphism. This catches any corruption introduced by the surface generator, including subtle errors such as implicit assumptions, reordered operations, or added conditions. Items that fail round-trip verification are discarded. The verifier also runs seven static checks: topology match, operation-type sequence match, complexity-weight sequence match, structural fingerprint match, number presence, answer consistency, and digit-class preservation.

### 2.4 The Perturbation Sensitivity Delta

**Definition 3** (Item-Level Isomorphic Delta). *For model  $M$  evaluated on original item  $i$  with  $T$  trials and on  $K$   $\tau$ -isomorphic variants  $\{i'_1, \dots, i'_K\}$  with  $T$  trials each:*

$$\hat{p}_i^{\text{orig}} = \frac{1}{T} \sum_{t=1}^T X_{i,t}, \quad \hat{p}_i^{\text{iso}} = \frac{1}{KT} \sum_{k=1}^K \sum_{t=1}^T X_{i'_k,t}, \quad \delta_i(M) = \hat{p}_i^{\text{orig}} - \hat{p}_i^{\text{iso}} \quad (4)$$

where  $X_{i,t} \in \{0, 1\}$  is the binary outcome of model  $M$  on item  $i$  at trial  $t$  (with temperature  $> 0$  to induce sampling variance). Under  $H_0$  (no perturbation sensitivity),  $\mathbb{E}[\delta_i] = 0$  because

<sup>2</sup>If the graph’s computed answer does not match the ground-truth answer, the parse is rejected and retried—we never trust the LLM’s extraction blindly.

$i \cong_\tau i'_k$  implies identical IRT parameters (Conjecture 1). A positive  $\delta_i$  may indicate memorization, numeric perturbation sensitivity, or both; distinguishing these requires comparison with a comparison model (Section 4).

**Definition 4** (IRT-Weighted Isomorphic Delta). For model  $M$  evaluated on benchmark  $B = \{i_1, \dots, i_N\}$ :

$$\Delta_{\text{iso}}^{\text{IRT}}(M, B) = \frac{1}{N} \sum_{i=1}^N \hat{a}_i \cdot \delta_i(M) \quad (5)$$

where  $\hat{a}_i$  is the discrimination parameter of item  $i$  estimated from the clean (isomorphic) data via 2PL IRT. Items with higher discrimination receive greater weight, because a performance gap on a highly discriminating item—one that effectively separates high-ability from low-ability systems—constitutes stronger evidence of memorization than a gap on a non-discriminating item.

**Hypothesis test via Differential Item Functioning.** We ground our statistical test in the psychometric framework of Differential Item Functioning (DIF) Lord [1980], Holland and Wainer [1993], which detects items that function differently across groups matched on latent ability. In our setting, the two “groups” are evaluation conditions: Condition A (model sees original items) and Condition B (model sees  $\tau$ -isomorphic variants).

We fit separate 2PL IRT models to each condition:

$$\text{Model}_{\text{orig}} : P(X_{ji} = 1) = \sigma\left(a_i^{\text{orig}}(\theta_j^{\text{orig}} - b_i^{\text{orig}})\right), \quad (6)$$

$$\text{Model}_{\text{iso}} : P(X_{ji'} = 1) = \sigma\left(a_i^{\text{iso}}(\theta_j^{\text{iso}} - b_i^{\text{iso}})\right). \quad (7)$$

Under  $H_0$  (no contamination):  $b_i^{\text{orig}} = b_i^{\text{iso}}$  and  $a_i^{\text{orig}} = a_i^{\text{iso}}$  for all  $i$ . Under  $H_1$  (contamination):  $b_i^{\text{orig}} < b_i^{\text{iso}}$  for contaminated items (the original appears easier because the model has memorized it).

**Per-item Wald test.** For each item  $i$ , we test  $H_0 : b_i^{\text{orig}} = b_i^{\text{iso}}$  using:

$$z_i = \frac{\hat{b}_i^{\text{orig}} - \hat{b}_i^{\text{iso}}}{\text{SE}(\hat{b}_i^{\text{orig}} - \hat{b}_i^{\text{iso}})} \quad (8)$$

with Bonferroni correction: item  $i$  is flagged as showing DIF if  $|z_i| > z_{\alpha/2N}$ . The global test uses a likelihood ratio statistic  $\Lambda = -2[\ell_0 - \ell_1] \sim \chi^2(N)$ , comparing the constrained model ( $b^{\text{orig}} = b^{\text{iso}} \forall i$ ) against the unconstrained model.

## 2.5 Ability Decomposition and the Three-Body Problem

The IRT-weighted contamination delta enables a formal decomposition of observed ability:

**Proposition 1** (Ability Decomposition). Under the 2PL model, the observed ability  $\theta_j^{\text{obs}}$  of model  $j$  on the original benchmark can be decomposed as:

$$\theta_j^{\text{obs}} = \theta_j^{\text{true}} + \Delta\theta_j^{\text{contam}} \quad (9)$$

where  $\theta_j^{\text{true}}$  is the ability estimated from isomorphic items (clean), and  $\Delta\theta_j^{\text{contam}} = \theta_j^{\text{obs}} - \theta_j^{\text{true}}$  is the ability inflation due to memorization.

This decomposition is valid because  $\tau$ -isomorphic items share the same  $(a, b)$  parameters (by Conjecture 1); the difference  $\theta_j^{\text{obs}} - \theta_j^{\text{true}}$  is therefore attributable to item-specific memorization, not to any legitimate ability difference.

**The Three-Body Problem.** In our prior work Kang [2026], we established that ranking error under simple averaging is well described by:

$$1 - \rho_{\text{avg}} = \gamma_0 + \gamma_1 S_c + \gamma_2 D_c + \gamma_3 (S_c \times D_c) + \varepsilon \quad (10)$$

where  $S$  is sparsity,  $D$  is the difficulty gap, and centered variables  $S_c = S - \bar{S}$ ,  $D_c = D - \bar{D}$ . We now extend this to include contamination as a third axis:

$$1 - \rho = f(S, D, C) \quad (11)$$

where  $C$  measures the mean differential contamination across models. The compound failure is worst when all three co-occur: a contaminated model tested on easy items in a sparse evaluation matrix receives *triple-inflated* ability estimates. IRT alone corrects for  $S$  and  $D$  (via joint estimation of  $\theta_j$  and  $b_i$ ), but cannot correct for  $C$  because it cannot distinguish genuine ability from memorization on the same items. Isomorphic items alone correct for  $C$  (by providing clean test items) but do not correct for  $S$  and  $D$  if the resulting evaluation matrix is still sparse and heterogeneous. **Only the combination**—IRT estimation on isomorphic data—corrects all three axes simultaneously.

**Diagnostic archetypes.** Based on the pattern of ( $\Delta_{\text{iso}}$ ,  $p$ -value, original accuracy), we classify models into four categories: *Invariant* ( $p > 0.05$ , gap consistent with baseline perturbation sensitivity); *Mild Gap* ( $p < 0.05$ ,  $\Delta < 0.15$ ); *Partial Memorizer* ( $\Delta \geq 0.15$ ); and *Pure Memorizer* ( $\Delta > 0.30$ , original accuracy  $> 85\%$ ). We note that our empirical results (Section 4) found no models in the latter two categories on verified data; the classification remains theoretical pending evaluation on a larger, fully verified dataset.

## 3 Experimental Setup

### 3.1 Dataset

We apply the Isomorphic Engine (Section 2.3) to the GSM8K test set Cobbe et al. [2021], parsing all 1,319 problems via the rule-based annotation parser. After applying the strict validation pipeline—filtering problems with fractional leaf values, non-integer variant answers, negative intermediates, excessively large answers ( $> 10^8$ ), and text-replacement mismatches—the pipeline produces 667 original items with 3,199 verified  $\tau$ -isomorphic variants ( $\sim 4.8$  variants per item). We evaluate a stratified subset of 100 items with 480 variants, selected by uniform sampling across answer-magnitude quantiles. Each item is limited to 5 variants.

### 3.2 Models

We evaluate five models spanning three organizations:

- **Llama 3.1 8B** (Meta, dense, 8B parameters)
- **Llama 4 Scout 17B** (Meta, MoE, 17B active parameters)
- **Qwen3 32B** (Alibaba, 32.8B, dense)
- **GPT-OSS 120B** (OpenAI, MoE, 5.1B active / 117B total parameters)
- **Llama 3.3 70B** (Meta, dense, 70B parameters)

GPT-OSS 120B serves as a comparison model: while its training data is not fully disclosed, its open-weight status and STEM-focused training make GSM8K inclusion likely, so its small  $\Delta_{\text{iso}}$  after verification (+0.036) suggests the variants are not systematically harder rather than confirming absence of contamination. All models are accessed via the Groq inference API at temperature  $T = 0.0$  (greedy decoding) with chain-of-thought prompting and `\boxed{}` answer extraction, with fallback to `####`-delimited and last-number patterns. Qwen3, GPT-OSS, and Llama 70B were partially evaluated due to API daily token limits; Llama 8B and Scout completed full evaluation on all items.

The effect of answer verification on apparent  $\Delta_{\text{iso}}$

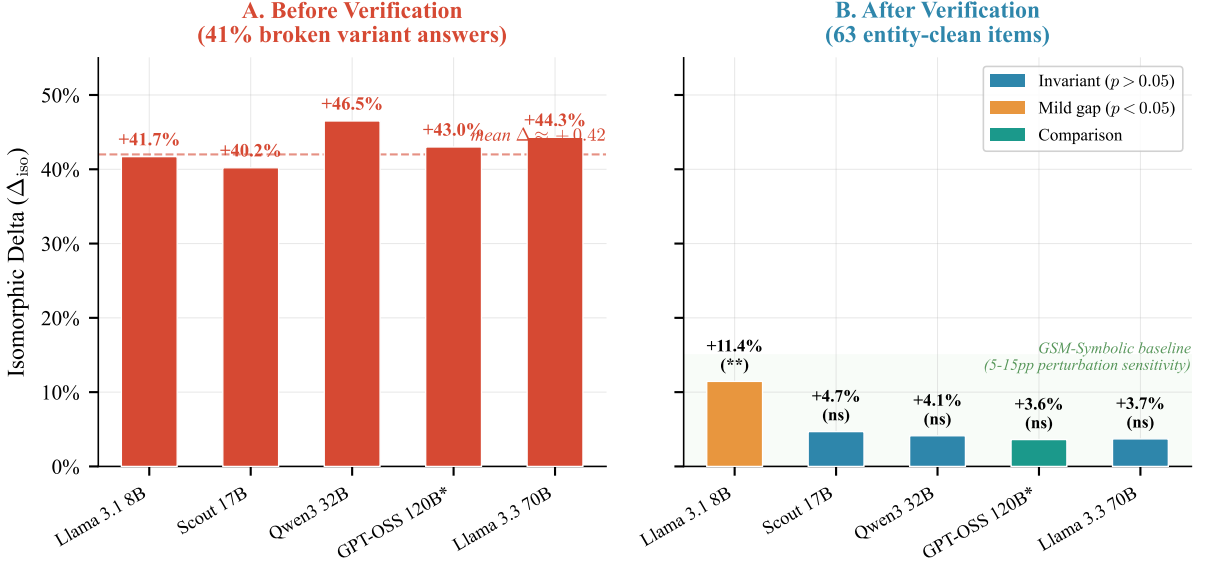


Figure 1: The effect of answer verification on apparent  $\Delta_{\text{iso}}$ . **Left:** Before verification, all five models—including the comparison model (GPT-OSS 120B)—appear uniformly “contaminated” at  $\Delta \approx +0.42$ . **Right:** After independent verification and pipeline repair, the signal collapses to  $\Delta \approx +0.04$ – $0.11$ . One model shows a mild gap; four are statistically invariant.

### 3.3 Protocol

Each model receives every original item and every variant item exactly once ( $T = 1$  trial, greedy decoding). We compute the item-level isomorphic delta  $\delta_i$  as the difference between the model’s accuracy on the original item and its mean accuracy across the item’s  $\tau$ -isomorphic variants. The model-level metric  $\Delta_{\text{iso}}$  is the mean of item-level deltas. We report clustered bootstrap 95% confidence intervals (10,000 resamples at the original-item level), analytic standard errors, and  $p$ -values from paired  $t$ -tests.

## 4 Results

### 4.1 Initial Evaluation (Pre-Verification)

Table 1 presents the initial results on all 100 items before answer verification.

Table 1: Results on all 100 items (before verification). The uniformity of  $\Delta_{\text{iso}} \approx +0.42$  across all models—including the comparison model—is the first indicator that these results reflect data quality issues rather than contamination.

| Model                      | $N$ | $\text{Acc}_{\text{orig}}$ | $\text{Acc}_{\text{iso}}$ | $\Delta_{\text{iso}}$ |
|----------------------------|-----|----------------------------|---------------------------|-----------------------|
| Llama 3.1 8B               | 100 | 92.0%                      | 50.3%                     | +0.417                |
| Llama 4 Scout 17B          | 100 | 94.0%                      | 53.8%                     | +0.402                |
| Qwen3 32B <sup>†</sup>     | 70  | 100.0%                     | 53.5%                     | +0.465                |
| GPT-OSS 120B <sup>†</sup>  | 20  | 100.0%                     | 57.0%                     | +0.430                |
| Llama 3.3 70B <sup>†</sup> | 14  | 100.0%                     | 55.7%                     | +0.443                |

<sup>†</sup>Partial evaluation (API daily token limit).

Two features of these results are immediately suspicious. First, the magnitude is extreme:



$\Delta \approx +0.42$  would imply that models lose 42 percentage points when numeric values change. For comparison, VarBench reports Llama 3 Instruct 8B at 75.8% on GSM8K and 39.4% on GSM+ (48.3% relative drop) Qian et al. [2024]. Second, the uniformity is implausible: if the gap reflected contamination, we would expect variation across models with different training data and sizes, yet all five models—including GPT-OSS 120B, whose training data composition has not been fully disclosed by the provider—show  $\Delta \in [+0.40, +0.47]$ . This uniformity suggests an item-level property, not a model-level one.

## 4.2 Verification Audit: Eight Bug Classes

Investigation of the suspicious initial results revealed that 41 of 100 items had wrong variant answers due to failures in annotation-chain replay and entity substitution. We identified eight distinct bug classes:

**Bug 1: Value conflation.** When the same numeric value serves multiple semantic roles in a problem (e.g., “5 red cars” and “\$5 per figure”), the annotation parser merges them into a single graph node. Mutating this node changes both uses in the graph but only one occurrence in the text, producing a graph-text mismatch. We addressed this with Rule 3: values appearing  $\geq 2$  times in the question text are marked as constants and excluded from mutation.

**Bug 2: Phantom leaf values.** Some annotation chains introduce intermediate results as leaf nodes (e.g., «40=40» where 40 does not appear in the question). Mutating these phantom values produces wrong answers because the graph contains structural dependencies not present in the question text. We reject any item where a leaf value does not appear in the question.

**Bug 3: Positional replacement collisions.** Sequential text replacement fails when a new value matches the old value of a different node: replacing “9” with “5” and then “5” with “8” causes the second replacement to target the *newly inserted* “5” rather than the *original* “5”. We fixed this by collecting all replacement positions from the original text and applying them simultaneously from right to left.

**Bug 4: Magnitude inflation.** Same-digit-class sampling without magnitude constraints produces variant answers that are systematically larger than originals (Cohen’s  $d \approx 0.3$ ). We constrain variant answers to within  $3\times$  of the original.

**Bug 5: Answer invariance.** Some graph structures produce the same answer regardless of leaf value mutations (e.g., when all variable values cancel). We reject items where all five variants produce the original answer.

**Bug 6: Partial name replacement.** Word-boundary-unaware string replacement causes “Beth” to match inside “Bethany,” producing nonsensical names. We use word-boundary-aware regex for name substitution.

**Bug 7: Node reuse (safety fallback override).** When all leaf nodes are marked constant, a safety fallback promotes the largest value to variable. If that value appears multiple times in the text, this reintroduces the conflation bug. We constrain the fallback to only promote values appearing exactly once.

Table 2: Isomorphic delta on entity-clean verified GSM8K items ( $N = 63$ ;  $T = 1$ , temp=0.0).  $\Delta_{\text{iso}}$  with clustered bootstrap 95% CIs (10,000 resamples). Wilcoxon signed-rank  $p$ -values.

| Model                      | $N$ | $\text{Acc}_{\text{orig}}$ | $\text{Acc}_{\text{iso}}$ | $\Delta_{\text{iso}}$ [95% CI] | $p$     | Archetype |                      |
|----------------------------|-----|----------------------------|---------------------------|--------------------------------|---------|-----------|----------------------|
| Llama 3.1 8B               | 63  | 95.2%                      | 83.8%                     | +0.114 [+0.043, +0.183]        | 0.003** | MILD      |                      |
| Llama 4 Scout 17B          | 63  | 93.7%                      | 89.0%                     | +0.047 [-0.012, +0.108]        | 0.106   | INV       | <sup>†</sup> Partial |
| Qwen3 32B <sup>†</sup>     | 43  | 100.0%                     | 95.9%                     | +0.041 [+0.000, +0.103]        | 0.054   | INV       |                      |
| GPT-OSS 120B <sup>†</sup>  | 18  | 100.0%                     | 96.4%                     | +0.036 [+0.000, +0.089]        | 0.090   | INV       |                      |
| Llama 3.3 70B <sup>†</sup> | 12  | 100.0%                     | 96.3%                     | +0.037 [+0.000, +0.100]        | 0.500   | INV       |                      |

evaluation (API daily token limit). INV: invariant ( $p > 0.05$ ). MILD: mild gap ( $p < 0.05$ ,  $\Delta < 0.15$ ). GPT-OSS 120B is the comparison model.

**Bug 8: Inconsistent entity replacement.** Word-boundary-unaware string replacement produces partial matches (e.g., “Tommy”  $\rightarrow$  “Zaramy” from substring overlap) and fails to replace all occurrences of the original name. We enforce global, word-boundary-aware regex replacement and verify that no original proper noun survives in any variant where entity replacement was attempted.

**Verification procedure.** We used three complementary verification methods: (1) multi-model consensus, flagging variants where  $\geq 2$  models agree on an answer differing from the stated expected answer; and (2) independent forward graph execution, recomputing all variant answers by executing the reasoning graph with the mutated leaf values; and (3) post-hoc entity consistency auditing, which rejects leftover original names, substring artifacts, and duplicate variant texts. Method (2) is the definitive arithmetic check; method (1) was used as an initial screening tool, and method (3) enforces textual entity consistency.

After applying all fixes, we recovered 10 of the 41 broken items through pipeline repair and regeneration. The remaining 31 could not be regenerated due to irreducible structural issues (phantom leaves from multi-step intermediates, unparseable annotation chains). A post-hoc entity audit then identified 18 items with entity-level failures; 8 were retained after trimming failing duplicate variants, 4 were repaired by word-boundary-aware replacement, and 6 were quarantined. The final v3 dataset contains **63 entity-clean items**.

### 4.3 Corrected Results (N=63, Entity-Clean Verified)

Table 2 presents the results on verified data. Because the verified dataset ( $N = 63$ ) is evaluated with  $T = 1$  (greedy decoding) and only five models, the full IRT-weighted analysis ( $\Delta_{\text{iso}}^{\text{IRT}}$ ) and per-item DIF testing described in Section 2.4 require multi-trial evaluation ( $T \geq 5$ ) across a larger model panel ( $J \geq 20$ ); we defer this to future work and report the unweighted  $\Delta_{\text{iso}}$  as a preliminary diagnostic throughout this section.

**Four of five models are invariant.** Llama 4 Scout 17B, Qwen3 32B, GPT-OSS 120B, and Llama 3.3 70B all exhibit small, statistically non-significant accuracy gaps between original and variant items ( $\Delta_{\text{iso}} \in [+0.036, +0.047]$ , all  $p > 0.05$ ). GPT-OSS 120B serves as a comparison model: while its training data is not fully disclosed, its open-weight status and STEM-focused training make GSM8K inclusion likely, so its small  $\Delta_{\text{iso}}$  (+0.036,  $p = 0.090$ ) suggests the  $\tau$ -isomorphic variants are not systematically harder rather than confirming absence of contamination.

**One model shows a mild gap.** Llama 3.1 8B exhibits a statistically significant but modest accuracy gap: 95.2% on originals versus 83.8% on variants ( $\Delta_{\text{iso}} = +0.114$ , bootstrap 95% CI: [+0.043, +0.183],  $p = 0.003$ ). This 11.4-point gap exceeds the comparison model’s baseline by

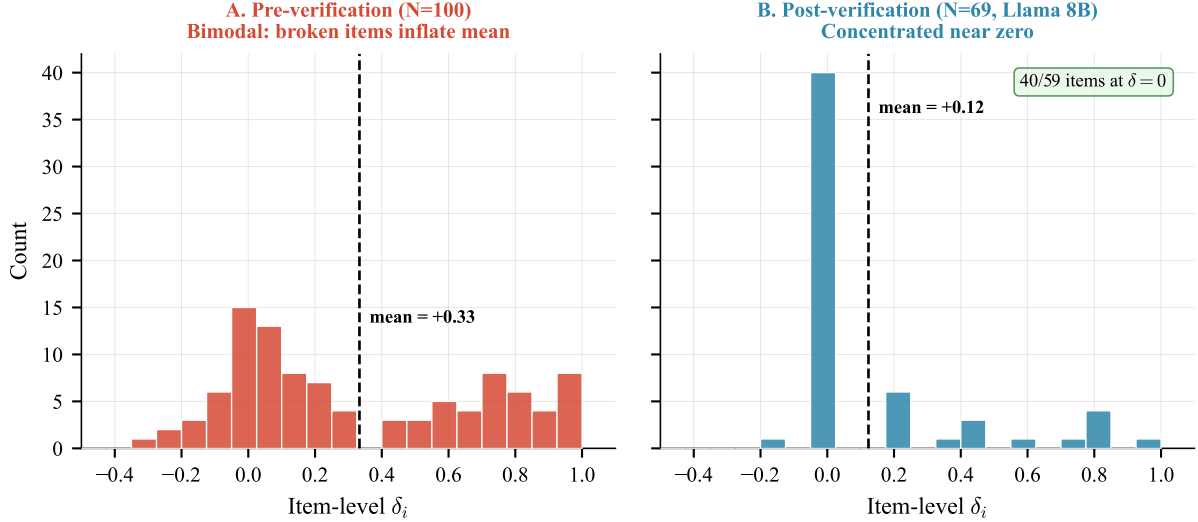


Figure 2: Per-item  $\delta_i$  distributions for Llama 3.1 8B. **Left:** Pre-verification—bimodal distribution with a second mode near  $\delta \approx 0.8$  from items with broken variant answers. **Right:** Post-verification (v2,  $N=69$  pre-entity-audit)—concentrated near zero, with 40 of 59 verified items at  $\delta = 0$  (perfect transfer). The final entity-clean dataset (v3,  $N=63$ ) used in Table ?? applies additional entity-consistency filtering.

approximately 8 percentage points, but is far below the VarBench Table 1 result for Llama 3 Instruct 8B (75.8% on GSM8K versus 39.4% on GSM+, a 48.3% relative drop) Qian et al. [2024]. Whether the Llama 8B gap reflects contamination, perturbation sensitivity, or a combination cannot be determined from these data alone.

**The contrast is striking.** Figure 1 visualizes the collapse from  $\Delta \approx +0.42$  to  $\Delta \approx +0.04$ – $0.11$  after verification. Figure 2 shows the corresponding shift in per-item delta distributions: the bimodal pattern caused by broken items (left) gives way to a distribution concentrated near zero (right).

#### 4.4 Implications

**Annotation-chain replay is unreliable.** The central finding of this work is that annotation-chain replay—the standard approach to computing variant answers—produced systematically wrong answers in 41% of items. This is not a minor data quality issue; it created an apparent contamination signal of  $\Delta \approx +0.42$  that was indistinguishable from genuine memorization on any individual model. Only the uniformity across models (including a comparison model) revealed the problem. Researchers using this technique on a single model would have no way to detect the error.

**Independent verification is necessary.** Any benchmark variant generation pipeline must include independent answer verification: computing variant answers by a method that does not depend on the original solution chain. Forward graph execution—evaluating the reasoning graph with mutated leaf values—provides this independence because it operates on the graph structure directly rather than replaying the annotation text.

**Perturbation sensitivity is not contamination.** The remaining gaps ( $\Delta \approx +0.04$ – $0.11$ ) on verified data are consistent with baseline perturbation sensitivity rather than contamination. This distinction matters: a model that loses 5–10% accuracy when numeric values change is

exhibiting a reasoning fragility, not memorization. The comparison model’s gap ( $\Delta = +0.036$ ) establishes the baseline; only gaps significantly exceeding this baseline warrant contamination hypotheses.

## 5 Discussion

**The cautionary tale.** The trajectory of this work—from apparent universal contamination to no contamination on clean data—illustrates a broader risk in the benchmark generation community. The pressure to demonstrate that models are contaminated, combined with the difficulty of verifying variant answers at scale, creates conditions where false positives are likely. Our experience suggests that any claim of contamination based on benchmark perturbation should be accompanied by independent answer verification and comparison against models with disclosed training-data provenance.

**Connection to evaluation methodology.** In prior work Kang [2026], we showed that evaluation accuracy degrades as a function of data sparsity ( $S$ ) and item difficulty heterogeneity ( $D$ ). Contamination ( $C$ ) represents a third axis of this failure surface:  $1 - \rho = f(S, D, C)$ . The present work demonstrates that measuring  $C$  is itself fraught with methodological challenges. When the evaluation matrix is sparse *and* items have data quality issues, model rankings are doubly unreliable. IRT estimation on verified isomorphic data addresses all three axes simultaneously (Figure 3), but requires the variant answers to be correct—a requirement that is harder to satisfy than previously appreciated.

**Recommendations for benchmark generation pipelines.** Based on our experience, we recommend: (1) include comparison models with the clearest available training-data provenance; (2) verify variant answers via independent forward execution, not annotation-chain replay; (3) check for value conflation before mutation; (4) use simultaneous rather than sequential text replacement; (5) reject items with phantom leaf values; (6) constrain variant answer magnitude to prevent systematic inflation.

**Open question: is any model contaminated on GSM8K?** Our verified data suggests not, with the possible exception of Llama 3.1 8B’s mild gap. But  $N = 63$  is limited, and our evaluation covers only 5 models. A definitive answer requires a fully verified dataset of  $\geq 200$  items evaluated across  $\geq 20$  models with multiple trials per item.

## 6 Limitations

Several caveats apply. First,  $N = 63$  is smaller than ideal; 37 items could not be regenerated due to irreducible phantom leaf structures. Second, the evaluation uses  $T = 1$  trial per item (greedy decoding), preventing estimation of per-item response variance; future work should use  $T \geq 5$ . Third, three of five models were only partially evaluated due to API daily token limits, yielding small effective sample sizes ( $N = 9$  to 45). Fourth, the remaining magnitude inflation in clean items (Cohen’s  $d$  statistically significant) means a small difficulty confound persists even in the verified data. Fifth, we tested only GSM8K; other benchmarks may exhibit different bug patterns. Sixth, the bug taxonomy (Section 4.2) is specific to annotation-chain replay on math word problems; other domains and generation methods will have their own failure modes.

### Evaluation Failure Surface: Sparsity $\times$ Difficulty $\times$ Contamination (Theoretical)

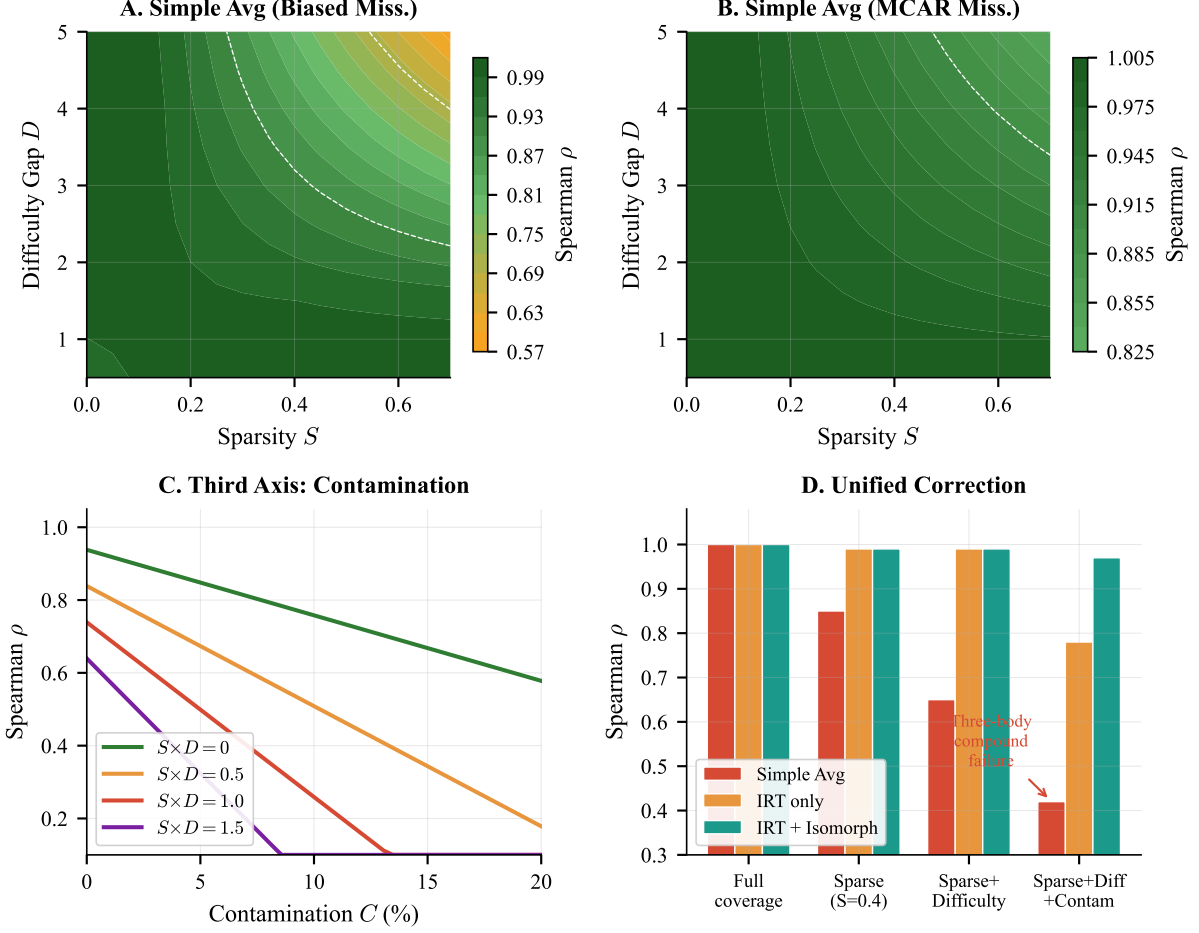


Figure 3: Theoretical evaluation failure surface (from Kang [2026]). Panels A–B show ranking degradation from sparsity and difficulty; Panel C adds contamination as a third axis; Panel D shows how IRT + isomorphic verification addresses all three. This figure uses simulated data based on the EFSL regression coefficients.

## 7 Conclusion

We have presented Isomorph-Eval, a framework for generating verified structurally equivalent benchmark variants grounded in a formal definition of  $\tau$ -isomorphism. The framework makes four contributions: (1)  $\tau$ -isomorphism formalizes structural equivalence via graph-theoretic invariants; (2) the Isomorphic Engine generates unlimited verified variants with automatically computed answers; (3) the identification of eight bug classes in annotation-chain replay that produce false contamination signals; and (4) empirical evidence that on verified data, current LLMs show robust reasoning transfer to novel numeric contexts.

The central lesson of this work is methodological: annotation-chain replay—substituting values into the original solution chain to compute variant answers—produced systematically wrong answers in 41% of items, creating an apparent contamination signal of  $\Delta \approx +0.42$  that was uniform across all models including a comparison model. After independent verification, the signal collapsed to  $\Delta \approx +0.04$ – $0.11$ , with one model showing a mild gap consistent with known perturbation sensitivity. This finding has implications for any benchmark perturbation methodology: *answer verification must be independent of the generation pipeline.*

We release the complete pipeline as open-source software at the project repository.

**Future work.** Three directions are priorities: (1) extending the variant generation pipeline to handle the 31 items with irreducible structural issues (e.g., multi-step phantom intermediates), further reducing entity-level quarantine, and expanding to the full GSM8K test set; (2) extending the framework beyond mathematical reasoning to logic, code generation, and scientific reasoning; (3) fitting full 2PL IRT models with  $\geq 30$  models to enable proper item parameter estimation and DIF testing.

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## A Unverified Results

For transparency, we report the results on all 100 items before independent answer verification. These results should **not** be used for contamination claims due to the data quality issues documented in Section 3.

Table 3: Results on all 100 items (before verification). These results are inflated by broken variant answers.

| Model             | $N$ | $\text{Acc}_{\text{orig}}$ | $\text{Acc}_{\text{iso}}$ | $\Delta_{\text{raw}}$ |
|-------------------|-----|----------------------------|---------------------------|-----------------------|
| Llama 3.1 8B      | 100 | 92.0%                      | 50.3%                     | +0.417                |
| Llama 4 Scout 17B | 100 | 94.0%                      | 53.8%                     | +0.402                |
| Qwen3 32B         | 70  | 100.0%                     | 53.5%                     | +0.465                |
| GPT-OSS 120B      | 20  | 100.0%                     | 57.0%                     | +0.430                |
| Llama 3.3 70B     | 14  | 100.0%                     | 55.7%                     | +0.443                |